

# What is Prosody and Prosodic Phonology? What is Prosodic Change?

Vertiefung Indoiranisch: Indoiranische und indogermanische Prosodie

17.10.2023

## Schedule (Tagesordnung)

### 1. Readings:

- This Week: Féry 2017: Ch. 1–3, Lahiri 2015, Nespor and Vogel 1986: Ch. 1–2, Byrd 2015: Ch. 1
- Next Week: Kager 1999: Ch. 1–3

2. Introduction: Core Problems and Questions
3. Prosody and Prosodic Phonology: Basic Definitions
4. Introducing the Prosodic Hierarchy
5. What is Intonation, Tone, Stress, and (Pitch)-Accent?
6. What is prosodic change?

## Hausmeisterthemen

### 1. Lehrplan: Kurs wird in ~4 Themenbereiche unterteilt.

- Prosodie im Allgemeinen und Analyse prosodischer Kategorien (Sitzungen 1–4)
- Allgemeines zum Akzent (Sitzungen 5–7)
- Akzent im Altindischen (Sitzung 8–11)
- Vergleichender Akzent der indogermanischen Sprachen (Sitzungen 12–15)

### 2. Lektüre

- Im Lehrplan mit Fettdruck gekennzeichneten Lektürestücke sind stark empfohlen als nötige Orientierung vor der jeweiligen Sitzung.
- Weitere Lektürestücke sind hilfreiche Zusätze, die zur Vertiefung des Themas dienen.
- Alle Lektüre befindet sich im per E-Mail geteilten Dropbox-Ordner unter "Lektüre".

### 3. Prüfung (VIS 2018)

- Klausur bei Frau Bozzone (Historische Grammar des Indoiranischen).
- Für diesen Kurs: Hausarbeit (44.000–56.000 Zeichen = ~ 12–16 Seiten).

- Thema: freie Auswahl im Bereich der indoiranischen / indogermanischen Phonologie (idealerweise zur Prosodie / Akzent)
- Abgabe der Hausarbeit: 20.03.2023, 11:59 Uhr (als PDF-Datei, per E-Mail an ryan.sandell@lrz.uni-muenchen.de / ryan.sandell@gmail.com).

## 1 Introduction: Core Problems and Questions

1. Consider the following facts, based on the data from Vedic Sanskrit (~ 1200–1000 BCE) and Classical Latin (from c. 100 BCE) below:
    - In the case of Vedic, the position of the prosodic prominence (I will call this STRESS) in each word (marked with an acute, the *udatta*) is seemingly arbitrary and unpredictable (“free stress”).
      - (1) Prosodic Prominence and Contrast in Vedic Sanskrit
        - a. *bráhmanasya* [brá:hmənəsja] ‘pious:M.GEN.SG’ vs. *brāhmanáśya* [bra:hmənəsja] ‘Brahman:M.GEN.SG’
        - b. *puróhitam* [puró:hitəm] ‘domestic priest:M.ACC.SG’ vs. *purútāmam* [puru:táməm] ‘much:MPV.M.ACC.SG’
        - c. *ápratidhr̥ṣṭaśavasam* [ə.prə.ti.d<sup>h̄</sup>ṛ.ṣṭə.çə.wə.səm] ‘having irresistible power:MASC.ACC.SG’ vs. *abhibhañj-at-ī-nám* [əb<sup>h̄</sup>ib<sup>h̄</sup>əñj-ət-i-ná:m] ‘smashing against-PART.ACT-FEM.-GEN.PL’
    - In the case of Latin, the position of stress is (with few exceptions and complications; see now Probert 2019 for detailed philological discussion) entirely predictable based on the number of syllables and the weight of the penultimate syllable (“fixed stress”; more on the use of parentheses and angled brackets here below and in coming weeks).
      - (2) Classical Latin Accent Assignment
        - a. Antepenult: nom.sg. (sá.pi.)⟨ēns⟩
        - b. Penult: nom.pl (sa.pi.)⟨én.⟩⟨tēs⟩
        - c. Antepenult, with unparsed penult (no degenerate feet permitted): cmpv.nom.sg. (sa.pi.)⟨én.⟩ti.⟨or⟩
        - d. **False:** cmpv.nom.sg. <sup>X</sup>(sa.pi.)⟨en.⟩⟨tí.or⟩
        - e. Penult: acc.sg. (ví).⟨rum⟩
        - f. Penult: vi(rúm)⟨que⟩
- ⇒ **Warning:** The two sentences above are replete with terms (“syllable”, “weight”, “stress”, etc.) that require definition, as well as an understanding of what those things are and how they work in a given language!
2. Patterns of both predictable and unpredictable stress are common across languages of the world. Unpredictable stress, leading to minimal pairs, occurs in many modern Indo-European languages, too, including Russian, Italian, and English.

- (3) Minimal Stress Pairs in Russian, Italian, and English<sup>1</sup>
- a. Russian (cf. further examples in Melvold 1989: 12)
    - i. *pláču* ['platʃu] 'weep:PRES.1SG'; *mýka* ['mukə] 'torture, pain'
    - ii. *plačú* [plə'tʃu] 'pay:PRES.1SG'; *myká* [mɔ'ka] 'flour'
  - b. Italian
    - i. *prìncipi* ['printʃipi] 'princes:PL'
    - ii. *principi* [prin'tʃi:pi] 'principles:PL'
  - c. English
    - i. *ímport* ['impɔ:t]
    - ii. *impórt* [ɪm'pɔ:t]
3. Simply determining how and why stress occurs where it does in the first case is not an easy problem in itself. Understanding the language-specific rules of where stress occurs within a word (more precisely, a prosodic word) is part of the study of a language's prosodic system.
4. Considering that both languages are closely related Indo-European languages attested in the 1st Millennium BCE, two yet more challenging questions pose themselves:
- A. **Reconstruction:** What did the prosodic grammar of the last common ancestor of these languages (Proto-Indo-European) look like?
  - B. **Language Change:** How and why did the prosodic grammars of languages like Vedic and Latin develop out of the prosodic grammar of PIE?
5. Methodological Principle: Synchronic analysis must precede diachronic analysis.
- It's fun (and sometimes enlightening) to try to formulate hypotheses concerning A. and B., but...
  - if our understanding of the synchronic grammars – where at least concrete data is available – is incomplete or incorrect, hypotheses about A. and B. will suffer accordingly.
6. The primary objective of this course is to apply a full array of tools from theoretical phonology to better understand the prosodic systems of ancient Indo-European languages.
- Primary *what*: assignment of stress at the level of the word.
  - Languages: focus on Vedic Sanskrit, but also Classical Sanskrit, Greek, Hittite, Germanic

## 2 Prosodic Phonology: Basic Definitions

7. The study of phonological processes can be conveniently divided into three levels, based on the size of the units with which the processes are concerned.
- **SEGMENTAL PHONOLOGY:** studies individual segmentable units that can be characterized as bundles of features (largely articulatory in nature).
  - **PROSODIC PHONOLOGY:** studies groupings of segments into larger units, as well as suprasegmental properties (tone, accent) that seem to stand “above” segments or may span grouping consisting of more than one segment. Also concerned with the phonological properties that constitute well-formed words in a language.

<sup>1</sup>Note that the use of acute and grave in these examples is not employed in the standard orthographies of these languages (either at all, or for the relevant lexemes, in the case of Italian). I have added these diacritics for the reader to signal the position of the primary stress.

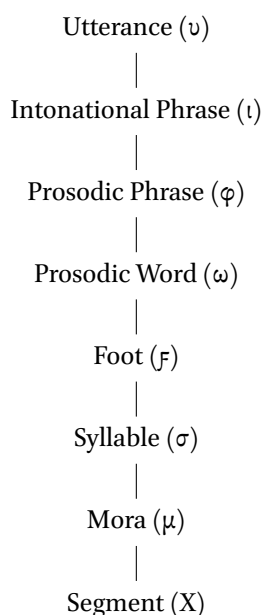
- INTONATIONAL PHONOLOGY: studies patterns of pitch that may span sequences of multiple words or directly depend upon the relationship between words.
8. Prosodic phonology and intonational phonology are often grouped together under the label of “suprasegmental phonology”, and not infrequently, the label “prosody” is used to encompass both prosodic phonology in the strict sense and intonational phonology.
  9. The usual targets of investigation for prosodic phonology are: **syllables** (groupings of segments), **feet** (groupings of syllables), **prosodic words** (lexical units normally consisting minimally of one syllable/foot), and **stress/accenuation** (determination of prosodic prominence internal to the prosodic word). Particularly the latter directly interfaces with the intonational phonology.
    - But important prosodic structures are often larger than the word (see below).
  10. Complication: Segmental phonological processes very frequently depend upon aspects of prosodic phonology.
    - The domains of syllable, foot, word, and phrase often determine where and when a process can apply or not (see below)
    - Suprasegmentals such as stress and tone may also condition or block the application of processes.
  11. On prosodic phonology in general, see Nespor and Vogel 1986, Itô 1988, Zec 1994, Peperkamp 1997, thorough recent introduction in Féry 2017.
  12. For a very introductory treatment of syllabification, stress assignment intonational phonology, see Hayes 2009: Ch. 13–15.
  13. Some historical and etymological background on “prosody”:
    - *prosody* < Gk. προσωδίᾱ [prosoːdíːa]
    - An explicit early definition of the lexeme is found in a fragment attributed to the Pre-Socratic philosopher Critias (Diels and Kranz 1952: Fragment 57:375–99): τὰς πρὸς κιθάραν ὠιδὰς προσωιδίας ἀρέσκει καλεῖν ‘it is suitable to call the προσωιδίας songs [ὠιδίας] proceeding from the κιθάρᾱ (a kind of lyre)’.<sup>2</sup>
    - Other later definitions (e.g., in Hesychius 3946; Latte 1953–1966) agree in considering “prosodies” to be songs with instrumental accompaniment. The form (acc.pl.) προσωδίας itself occurs once in the Platonic corpus (*Republic* 399a8), but appears extensively in the Aristotelian corpus (11 ×) and thereafter appears as a common term in works of or related to the Alexandrian grammarians.
    - The connection of προσωδίᾱ with ᾠδή [ɔːdíː] ‘song’ (borrowed into English as *ode*), in turn a derivative of the verb αἰδῶ [aeíɔː] ‘sing’ (earlier form ἀοιδή [aoídeː], without vowel contraction, mostly attested in Homer), is etymologically transparent, and indeed then προσωδίᾱ is correctly to be regarded as a type of song or singing.
    - Literally, it is a ‘singing’ (ὠδία [ɔːdíːa]) ‘against’ (πρός [prós] < PIE \*[próti]; cf. Skt. *práti* ‘to, against’). This etymological transparency also permitted a literal translation of the term into Latin as *accen-tus* (< *ad* ‘to, towards’ + *canere* ‘to sing’, in the form of a past passive participle); cf. Probert 2003: 3 and Allen 1987: 116.

<sup>2</sup>One attested occurrence which is likely earlier (probably early 5. c. BCE) is in a fragment of tragedy attributed to Aeschylus; see Radt 1985: Fragment 299.

### 3 Introducing the Prosodic Hierarchy and Prosody-sensitive Rules

14. The fundamental empirical difficulty posed by the investigation into prosodic structure, i.e., the suprasegmental phonological organization of speech, is that its units are, for the most part, in themselves imperceptible and must therefore be examined indirectly through the effects that they exercise.
15. Since at least Selkirk 1984 and Nespor and Vogel 1986, most phonologists have assumed a “hierarchy” of prosodic categories that stack on top of and encompass units of the lower category.

(4) Schematic PROSODIC HIERARCHY



(5) Definitions of Constituents in the PROSODIC HIERARCHY (cf. Féry 2017: 36)<sup>3</sup>

| Abbreviation   | Prosodic Constituent         | Definition (~ Syntactic Equivalent)                           |
|----------------|------------------------------|---------------------------------------------------------------|
| υ (U)          | Utterance                    | Complete expression without pause, including multiple clauses |
| ι (I)          | Intonational Phrase          | Clause (multiple XPs)                                         |
| φ (Φ)          | Phonological/Prosodic Phrase | Phrase (XP)                                                   |
| ω              | Prosodic Word                | Grammatical Word (X)                                          |
| ƒ (φ, F, f, Σ) | Foot                         | Metrical Unit (e.g., trochee)                                 |
| σ (Σ)          | Syllable                     | Segmental string (e.g., CVC)                                  |
| μ              | Mora                         | Timing unit / syllable weight                                 |
| X/C/V          | Segment                      | Concrete syllable nucleus (V) or margin (C)                   |

16. As a concrete example, consider the following sentence of German. I assume that function words in this example, including disyllabic ones and sequences of multiple function words, are clitics parsed

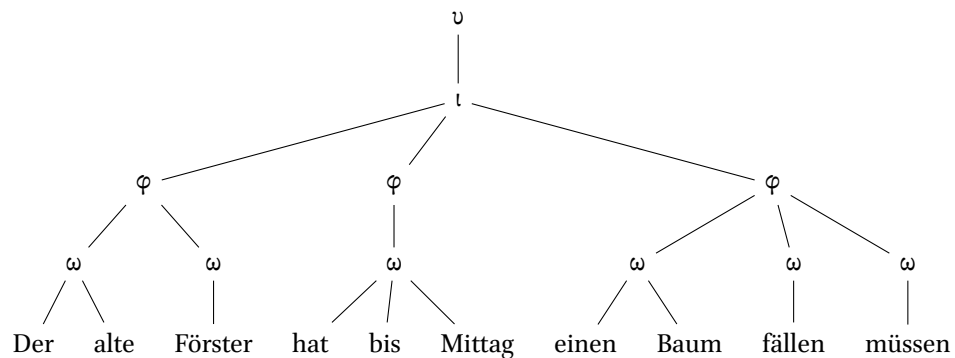
<sup>3</sup>Practice in the abbreviation conventions for the prosodic hierarchy varies; almost universally employed are  $\sigma$  (syllable) and  $\omega$  (prosodic word), though for the form one sometimes encounters the uppercase sigma,  $\Sigma$ . The primary symbol shown here will be used throughout this work; the alternative in parentheses is also encountered reasonably often. The labels here follow Féry 2017, with the exception of the foot, for which I introduce the sign lowercase digamma ( $\mathfrak{f}$ ), in order to consistently apply lowercase Greek letters for all constituents of the prosodic hierarchy (note that the Greek letter digamma, while not part of the familiar Ionic alphabet, is the source for Latin ⟨F⟩). Compare further Selkirk 2011 or Itô and Mester 2013: 26 on the correspondences between prosodic categories and syntactic categories.

into prosodic words with neighboring content words, though variation may occur based on e.g., speech style. See further Itô and Mester 2009 for specific arguments in favor of prosodic-word adjunction in German.

(6) Prosodic Structure in German

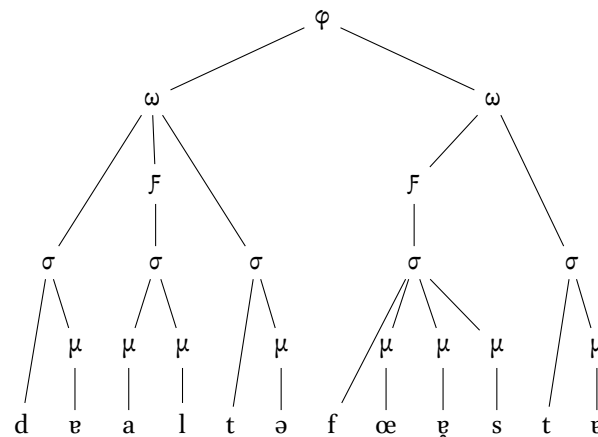
- a. Der alte Förster hat bis Mittag einen Baum fällen müssen.  
 the.NOM.M old forester.M have.PRES.3SG until noon a.ACC.M tree.M fell.INF must.PTCP.PST  
 ‘The old forester had to (= *hat müssen*) fell a tree by noon.’

b. Prosodic Structure at  $\omega$  and above

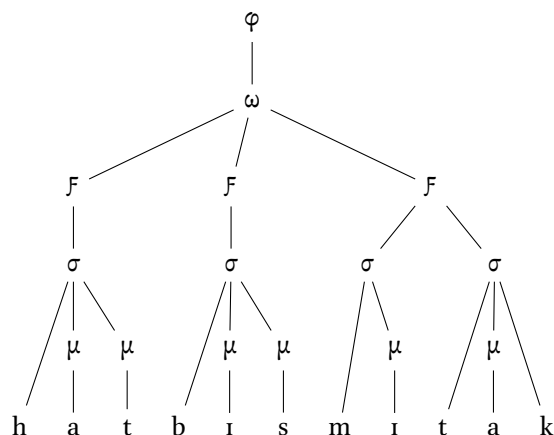
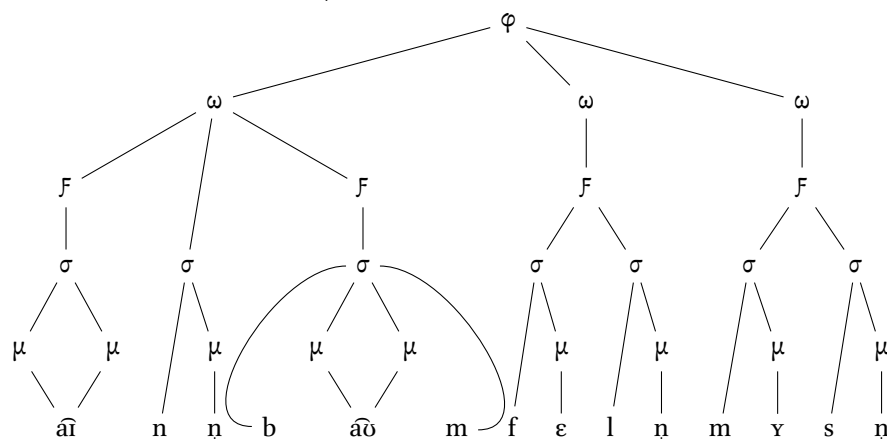


c. Prosodic Structure from  $\omega$  and below, by  $\varphi^4$

i. [Der alte Förster] $_{\varphi}$



<sup>4</sup>I assume non-exhaustive parsing of prosodic words into feet, constructing strictly bimoraic feet, and assume the extrametricality of word-final consonants. See further Alber 1998 for an analysis of foot construction in German.

ii. [hat bis Mittag]<sub>φ</sub>iii. [einen Baum fällen müssen]<sub>φ</sub>

17. According to the conception of the Prosodic Hierarchy advanced by Nespor & Vogel, well-formed prosodic structures are subject to a set of four specific design principles (laid out in Nespor and Vogel 1986: 7), the first two of which are frequently summarized under the label of the STRICT LAYER HYPOTHESIS (though this term originates with Selkirk 1984: 26; cf. also Inkelas 1990: §2.2, Itô and Mester 2009: 146, or Selkirk 2011: 437). Nespor & Vogel's Principle 1 and 2 are reproduced here, followed by the version of the Strict Layer Hypothesis stated in Selkirk 2011.

- (7) Principles 1 and 2 of Prosodic Structure Construction (Nespor and Vogel 1986: 7)

*Principle 1.* A given nonterminal unit of the prosodic hierarchy,  $X^P$ , is composed of one or more units of the immediately lower category,  $X^{P-1}$ .

*Principle 2.* A unit of a given level of the hierarchy is exhaustively contained in the superordinate unit of which it is a part.

- (8) The STRICT LAYER HYPOTHESIS (Selkirk 2011: 437)

A constituent of category-level  $n$  in the prosodic hierarchy immediately dominates only a (sequence of) constituents at category-level  $n-1$  in the hierarchy.

- This conception, and the STRICT LAYER HYPOTHESIS, is now widely believed to be too strong.
- Most phonologists now probably believe 1) that “skipping levels” is allowed, and 2) that the categories of  $\omega$  and  $\phi$ , at least, and occur recursively.

18. These prosodic domains are also frequently responsible for conditioning the application of segmental changes.

(9) Prosodic Domain Rules

- a. INTERVOCALIC SPIRANTIZATION in Tuscan Italian (cf. Nespor and Vogel 1986: 209)

$$\begin{bmatrix} +\text{delayed release} \\ -\text{continuant} \end{bmatrix} \rightarrow [+continuant] / [\dots[-consonantal] \_\_\_ [-consonantal] \dots]_i$$

Affricates are realized as corresponding fricatives between  $[-consonantal]$  segments contained within the same intonational phrase.

/[il mio kriʃeto tʃerka il suo ʃibo neʎi angoli del:a gab:ja]\_i/  
 → [il mio kriʃeto ʃerka il suo ʃibo neʎi angoli del:a gab:ja]\_i

‘my hamster looks for its food in the corners of the cage’

- b. OBSTRUENT VOICING in Sanskrit

$$[-\text{sonorant}] \rightarrow [+voice] / [\dots \_\_\_]_\omega [[+voice] \dots]_\omega \dots]_\upsilon$$

An obstruent is realized as  $[+voice]$  if it occurs at the right edge of a prosodic word, the initial segment of the following prosodic word is  $[+voice]$ , and both prosodic words are contained within the domain of the same utterance.

- c. *visarga*-FORMATION in Sanskrit

$$\begin{Bmatrix} s \\ r \end{Bmatrix} \rightarrow [h] / \_\_\_]_\upsilon$$

A segment  $/s/$  or  $/r/$  is realized as  $[h]$  in absolute utterance-final position.

## 4 What is Intonation, Tone, Stress, and Accent?

19. “Accent” or “Prosodic Prominence” intuitively holds over the domain of the syllable, or is at least a property of the nucleus of the syllable.
20. While prosodic prominence has direct phonetic correlates (in the form of pitch, intensity, and/or duration) groupings of segments in syllables and syllables into feet have no direct phonetic correlates.
21. Rough and approximative definition of “accent”: some syllable or segment internal to a prosodic word that can be identified as more “prominent” on the basis some phonetic properties.
- The relevant phonetic properties (some combination of) greater duration, higher amplitude levels (“intensity”), and greater fundamental frequency (higher pitch, raised  $F_0$ ).
  - Accent can then be thought of as fundamentally a RELATIONAL property.
  - Insofar as “accent” serves to pick out “prominent” constituents internal to prosodic words internal to a given language, its precise phonetic cues are not crucial. The particular cues indicative of prominence may thus vary cross-linguistically.
22. “Stress accent language” vs. “pitch accent language” vs. “tonal language” (cf. generally Yip 2002). Traditional distinctions:
22. 1. Pitch accent: prominence marked principally by higher pitch (e.g., Japanese, Vedic, Ancient Greek).
22. 2. Stress accent: prominence marked by higher amplitude and duration, and usually higher pitch as well (e.g., English, Italian, Russian, Modern Greek).

22. 3. Tonal: lexical pitch contrasts within and between words (e.g., Navajo, Mandarin, Vietnamese). Prominence selection (“stress”) may or may not be present.
- The first two types reflect different possible phonetic realizations of a unit that is determined to be “accented”, under both traditional views (e.g., Hirt 1895) and some current approaches to the analysis of stress and accent (e.g., van der Hulst 2009, van der Hulst 2012).
23. In “tonal” languages, each possible tone bearing unit may be either lexically specified for some tonal properties, receive a “default” tone, or be assigned a tone by some phonological processes.
- Crucially, in such languages no segment or syllable is identifiably more prominent than any other segment/syllable.
  - In Navajo, all vowels are either specified as [+ high tone] or receive a default low tone. Rising and falling tones may arise from vowel contraction across morphemes. The number of high and low tones in a phonological word is unlimited.
24. Larry Hyman has, on the other hand, argued in a series of publications, that “pitch accent” is not a meaningful category, being used to describe many languages with many distinct types of prosodic systems (Hyman 2006, Hyman 2009, Hyman 2014).
- From Hyman’s perspective, languages that conform to the prototype of “stress” all exhibit the following phonological properties of word-level prosody: CULMINATIVITY, OBLIGATORINESS, SYLLABLE-ORIENTATION, PRIVATIVITY, and METRICITY.
- (10) Minimum Properties of Languages with Stress (after Hyman 2014: 60–1)
- a. SYLLABLE-DESIGNATION: the stress-bearing unit is the syllable (cf. Hayes 1995: 49–50)
  - b. OBLIGATORINESS: every lexical word has at least one syllable marked for the highest degree of metrical prominence (= primary stress).
  - c. CULMINATIVITY: every lexical word has at most one syllable marked for the highest degree of metrical prominence.
  - d. PRIVATIVITY: a given a given foot/syllable/mora either has prominence or lacks it, such that the prosodic constituent may be labeled either “stressed” or “unstressed”. Pitch may similarly be treated privatively, if TBUs need only be specified as bearing one type of pitch, e.g., only an underlying contrast between /H/ or /Ø/.
  - e. METRICITY: prosodic prominence is positionally restricted (typically by reference to foot structure).
- Some languages may lack one or more of these properties. Such languages are frequently given the label “pitch-accent language”, though they may differ in crucial respects.
  - Tōkyō Japanese lacks OBLIGATORINESS.
  - Seneca (Iroquoian) lacks both OBLIGATORINESS and CULMINATIVITY, though the assignment of high tone seems dependent on foot structure.
  - Note that Vedic Sanskrit and Ancient Greek seem very close to Hyman’s phonological stress prototype.
25. Some languages may have both a clear “prominence” *and* contrastive tonal contours.
- In other words, these are languages with both stress and tone.
  - This situation is characteristic of Balto-Slavic languages, both historically and into the present day.

- Latvian has a phonologically predictable prominence (initial syllable), but contrastive tonal contours (always realized in the initial syllable; not always realized in all dialects in non-initial syllables).
  - Many dialects of Lithuanian, Serbo-Croatian, and Slovene have contrastive tones which usually are only realized under the (phonologically unpredictable) accent.
  - See the overviews in Petit 2010 and Jasanoff 2017 on Baltic and Slavic.
  - The same description can be applied to some varieties of North Germanic (e.g., Stockholm Swedish) and West Germanic (e.g., southern Dutch dialects): the languages have a lexical distinction in tone, as well as stress. All words have stress, some lexical items have a lexical tone, with the result that different lexemes show different tones. See Riad 2014 for details on the Swedish system.
26. When asking questions about the accentuation of old Indo-European languages, or their common proto-language, two questions must be answered:
- 26.1. what are the phonetic correlates of the most prominent units in a prosodic word (i.e., “stress”, “pitch”, neither)?
- 26.2. how is the most prominent unit within a prosodic word determined?
27. We will discuss the overall typology of word prosody and the analysis of predictable stress systems in much greater detail in **Sitzung 5**.

## 5 What is prosodic change?

28. Most broadly, a prosodic change could be characterized as any change to the phonological system above the level of the segment! E.g.:
- rules for constructing syllables;
  - rules for assigning stress;
  - the substance of particular intonational contours (e.g., the tune associated with information-seeking questions).
29. Our focus in this course will largely be concerned with the rules and constraints that govern that assignment of **stress**, so I will present a handful of examples here.
30. As seen above, the position of stress in Vedic Sanskrit was unpredictable.
- Within the Vedic period, some words and lexical categories more generally exhibit a change in the position of stress.
  - For example, abstract nouns in *-ti-* exhibit a tendency to change from having stress on the suffix, *-tí-*, to having stress on the root.
- (11) Changes in Vedic *-ti*-Abstracts
- a. *matí-* ‘mind’ (RV) > *máti-* (ŚB) – see Lundquist 2015
  - b. *śaktí-* ‘power, ability’ (RV Family Books) > *śákti-* – see Sandell 2016
- One possible account: stress assignment in Vedic depends on morphological structure, so morphologically complex and morphologically simplex words might have different stress:
  - Complex /ma-ti-/ → *matí-*; simplex /mati-/ → *máti-*.

- See Probert 2006 on this idea in general, with specific application to Greek.
- Problem with this account: morphologically simplex disyllabic words in Vedic can have stress on the first (*prāti* ‘towards, against’) or second (*abhí* ‘to, towards’) syllable. So why change the stress at all?

31. Diachronically, the notion of PERTINACITY in prosody argued for in Lahiri 2015 looks largely valid.

- “Pertinacity”, in so many words, just means tendency to analyze the same data in the same fashion.
- Pertinacity can exhibit itself in one of two ways:
  1. “Same outputs, different grammar”: some aspect of the grammar changes, but the surface output in a subset of the outputs remains unchanged.
  2. “Different outputs, same grammar”: the surface output in a subset of all outputs changes, even though a given aspect of the grammar remains unchanged.
- Lahiri (2015: 225–6) nicely illustrates how different sets of forms in a hypothetical Pre-Latin > Classical Latin stress grammar illustrate the first principle.
  - Parametric formulations of stress assignment for Pre- and Classical Latin (see also Nishimura 2011, Nishimura 2014 for overviews of stress in the history of Italic and Latin):

(12) Pre-Latin

- a. *Main Left*
- b. Moraic Trochee (foot structures of the shapes ( $\bar{\sigma}$ ) or ( $\check{\sigma} \check{\sigma}$ ))
- c. Right-to-Left parsing
- d. Final Syllable Extrametricality

(13) Classical Latin

- a. *Main Right*
- b. Moraic Trochee
- c. Right-to-Left parsing
- d. Final Syllable Extrametricality

- In all four possible permutations of trisyllabic words, only /H H X/ will have more than one foot (assuming no type of degenerate feet), and thus only words of that shape will show stress in a different position, given a change in the edge alignment of main stress.
- Extending Lahiri’s observation to quadrisyllabic words, we can make the comparison below, which illustrate that such longer words do tend to turn out very differently.

(14) Stress Comparison of Quadrisyllabic Words in Pre- and Classical Latin

| Pre-Latin          | Classical Latin    | Changed Output? |
|--------------------|--------------------|-----------------|
| [L (‘L L) ⟨X⟩]     | [L (‘L L) ⟨X⟩]     | Same            |
| [(‘L L) (H) ⟨X⟩]   | [(L L) (‘H) ⟨X⟩]   | Different       |
| [L (‘H) L ⟨X⟩]     | [L (‘H) L ⟨X⟩]     | Same            |
| [L (‘H) (H) ⟨X⟩]   | [L (H) (‘H) ⟨X⟩]   | Different       |
| [(‘H) (H) (H) ⟨X⟩] | [(H) (H) (‘H) ⟨X⟩] | Different       |
| [(‘H) (H) L ⟨X⟩]   | [(H) (‘H) L ⟨X⟩]   | Different       |
| [(‘H) (L L) ⟨X⟩]   | [(H) (‘L L) ⟨X⟩]   | Different       |
| [(‘H) L (H) ⟨X⟩]   | [(H) L (‘H) ⟨X⟩]   | Different       |

- In words with 5 or more syllables, there will always be at least two feet; the C. Lat. outputs will thus always be different from the Pre-Lat. outputs.

- Statistical note: in a small Latin corpus (73000 tokens, Caesar)  $\sim 71\%$  of tokens have three or fewer syllables, another  $\sim 19\%$  have four syllables. Among trisyllabic forms, different word shapes are approximately equally distributed.
  - $\Rightarrow$  Rough estimate:  $\sim 25 - 33\%$  of Latin word tokens would be disambiguating between a Main Left and a Main Right Grammar.
  - ☞ Is it credible to believe that the evidence for Main Left would have been insufficient to acquire it, and instead to posit Main Right?
- How many outputs result in a CLASH is a secondary stress is assigned to each foot? The trisyllabic  $[('H) (H) \langle X \rangle]$ , plus 4/8 quadrisyllabic patterns. Thus,  $\sim 15\%$  of all tokens would be different and have stress clash (ignoring words longer than 4 syllables).
- As a further example of pertinacity, consider the statement of Roca (1999: 769–70) (to be agreed with in spirit, if not letter of analysis):
 

Romance nominal foot structure can easily be traced back to Latin, of which language it keeps the left-headed binary foot built from the right, the extrametricality of the last syllable (confined, however, to the desinential constituent, or else restricted to a handful of lexically marked items), and the accenting of the stem's final element (light syllables included in all but exceptional forms). At the word level, the identity between the Latin and Romance algorithms is total: right-headed constituent construction followed by line 1 conflation.
- As we will see in several instances throughout the course, pertinacity indeed seems to be the normal state.
- Motivating and truly understanding *why* prosodic change occurs is hard.

## 6 Thanks!

32. **Next Week:** Understanding Optimality Theory and constructing prosodic structures with constraints

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